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(a) (i) $v = 2\pi r/T$, ($= 2\pi \times 6.2/4.1$) [1]
 $= 9.5 \text{ (m s}^{-1}\text{)}$ (9.50) [1] [2]

(ii) acceleration $= v^2/r$ ($= 9.502/6.2$) [1]
 $= 14.6 \text{ (m s}^{-2}\text{)}$ [1] [2]

(b) (i) mg unaltered (beyond arms but not beyond bottom of cage) $mv^2/r > mg$ [1]
 contact force downward smaller than weight [1] [2]

(ii) $R + mg = ma$, $R = m(a - g)$ [1]
 $= 75(14.6 - 9.81) = 360 \text{ (N)}$ [1] [2]

(c) (i) $R = ma - mg$, $R > 0$ is required to remain contact [1]
 so $a > g$, i.e. $v > \sqrt{rg}$, the minimum speed for a contact force to be required [1] [2]

(ii) minimum speed when $v^2/r = g$ } [1]
 $v^2/6.2 = 9.81$ }
 $v = 7.8 \text{ (m s}^{-1}\text{)}$ [1] [2]

[12]

Comments: The magnitude of centripetal force acting on the man is constant, so in (b) (i) the length of normal contact force at A should be drawn shorter than the length of weight, so that the length of the vector sum is similar to that when the man is at B. In (c) (ii), some used the expression $v\omega$ for the centripetal acceleration and calculated the value of v incorrectly as $g/\omega = 9.81/1.53 = 6.4 \text{ m s}^{-1}$.

3 (a) (i) gravitational force provides centripetal force

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